# E-commerce Data Analysis

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## Problem Statement

### Project Description:

This Project focuses on analyzing e-commerce data using SQL to answer business-related questions and generate meaningful insights.

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Business Questions

Question-1: You can analyze all the tables by describing their contents.

Task: Describe the Tables:

1. Customers
2. Products
3. Orders
4. OrderDetails

Question-2: Identify the top 3 cities with the highest number of customers to determine key markets for targeted marketing and logistic optimization.

Hint:

Use the “Customers” Table.

Return the result table limited to top 3 locations in descending order.

a) As per this query's result, Which of the cities must be focused as a part of marketing strategies?

Question-3: Determine how many customers fall into each order frequency category based on the number of orders they have placed.

Using the Orders table, calculate the number of customers who placed 1 order, 2 orders, 3 orders, etc.

Return a table showing:

The number of orders placed

The count of customers who placed that many orders

Sort the results by NumberOfOrders in ascending order.

a) As per the Engagement Depth Analysis question, What is the trend of the number of customers v/s number of orders?

b) As per the Engagement Depth Analysis question, Which customer category does the company experience the most?

Question-4: Identify products where the average purchase quantity per order is 2 but with a high total revenue, suggesting premium product trends.

Hint:

Use “OrderDetails”.

Return the result table which includes average quantity and the total revenue in descending order.

a) Among products with an average purchase quantity of two, which ones exhibit the highest total revenue?

Question-5: For each product category, calculate the unique number of customers purchasing from it. This will help understand which categories have wider appeal across the customer base.
Hint:
Use the “Products”, “OrderDetails” and “Orders” table.
Return the result table which will help you count the unique number of customers in descending order.
a) As per this question, Which product category needs more focus as it is in high demand among the customers?

Question-6: Analyze the month-on-month percentage change in total sales to identify growth trends.
Hint:
Use the “Orders” table.
Return the result table which will help you get the month (YYYY-MM), Total Sales and Percent Change of the total amount (Present month value- Previous month value/ Previous month value)*100.
The resulting change in percentage should be rounded to 2 decimal places.
a) As per Sales Trend Analysis question, During which month did the sales experience the largest decline?
b) As per Sales Trend Analysis question, What could be inferred about the sales trend from March to August?

Question-7: Examine how the average order value changes month-on-month. Insights can guide pricing and promotional strategies to enhance order value.
Hint:
Use the “Orders” Table.
Return the result table which will help you get the month (YYYY-MM), Average order value and Change in the average order value (Present month value- Previous month value).
Round the average order value to two decimal places before calculating the month-on-month change. Also the resulting AvgOrderValue and ChangeInValue columns should both be rounded to two decimal places, and the final results should be ordered in descending order of ChangeInValue.
a) As per last question, Which month has the highest change in the average order value?

Question-8: Based on sales data, identify products with the fastest turnover rates, suggesting high demand and the need for frequent restocking.
Hint:
Use the “OrderDetails” table.
Return the result table limited to top 5 products according to the SalesFrequency column in descending order.
a) As per this question, Which product_id has the highest turnover rates and needs to be restocked frequently?

Question-9: List products purchased by less than 40% of the customer base, indicating potential mismatches between inventory and customer interest.
Hint:
Use the “Products”, “Orders”, “OrderDetails” and “Customers” table.

Return the result table which will help you get the product names along with the count of unique customers who belong to the lower 40% of the customer pool.
a) Why might certain products have purchase rates below 40% of the total customer base?
b) After running an analysis to identify products purchased by less than 40% of the customer base, it was found that a few products have lower purchase rates than expected.
What could be a strategic action to improve the sales of these underperforming products?

Question-10: Evaluate the month-on-month growth rate in the customer base to understand the effectiveness of marketing campaigns and market expansion efforts.
Hint:
Use the “Orders” table.
Return the result table which will help you get the count of the number of customers who made the first purchase on a monthly basis.
The resulting table should be ascendingly ordered according to the month.
a) As per this question, What can be inferred about the growth trend in the customer base from the result table?

Question-11: Identify the months with the highest sales volume, aiding in planning for stock levels, marketing efforts, and staffing in anticipation of peak demand periods.
Hint:
Use the “Orders” table.
Return the result table which will help you get the month (YYYY-MM) and the Total sales made by the company limiting to the top 3 months.
The resulting table should be in descending order suggesting the highest sales month.
a) As per this question, Which months will require major restocking of product and increased staff?